

Substation ESG Report

Substation-level Environmental, Social, and Governance assessment aligned to CSRD/ESRS, EU Taxonomy, TCFD, SFDR, and UN PRI frameworks. Updated annually via the SSI Index scoring engine.

SUBSTATION

ICE - Central Telefónica

w1230905408 · Heredia, heredia

SSI SCORE (R_{MEDIAN})

0.586

● **High Risk** 85.8th percentile

VOLTAGE CLASS

230 kV

High-voltage transmission

CONFIDENCE INTERVAL

0.536 – 0.646

P5–P95 · CI width 0.110 · Medium-High confidence

MARKOV ETTC

35 yr

Expected time to criticality

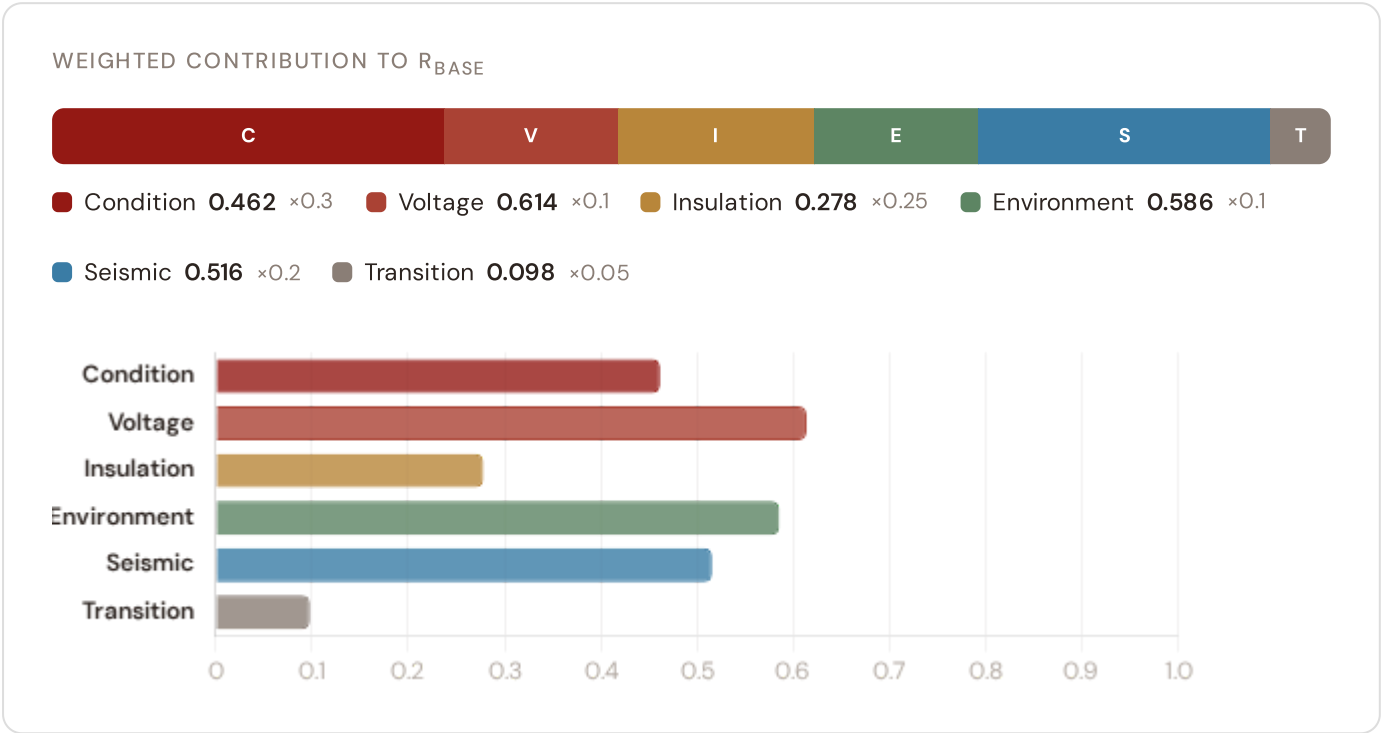
DER PENETRATION

0.06

T1 transition stress score

Component Fingerprint

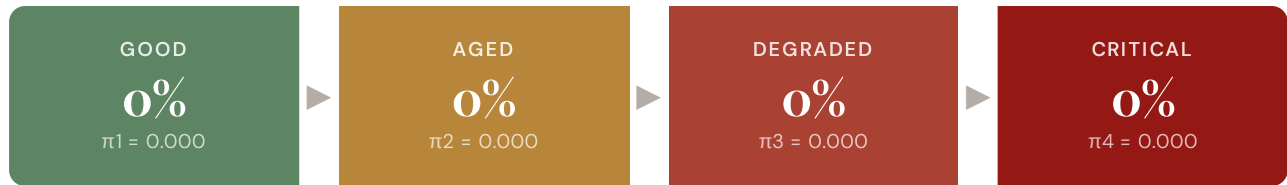
Six-component weighted decomposition of the SSI composite score — the substation's structural risk profile



Markov Degradation Profile

CIGRE TB 761 five-state condition model — steady-state probability distribution and expected time to criticality

STEADY-STATE DISTRIBUTION



Markov 5-state model calibrated from CIGRE Technical Brochure 761 (2019). States represent asset condition from new/good through critical. Steady-state probabilities indicate the long-run proportion of time the asset spends in each condition state given current operating environment.

RISK SCORE

58.600

Composite Markov risk

ETTC

35 yr

Expected time to critical

P(CRITICAL) 20YR

35.2%

Probability of reaching critical

CORROSION

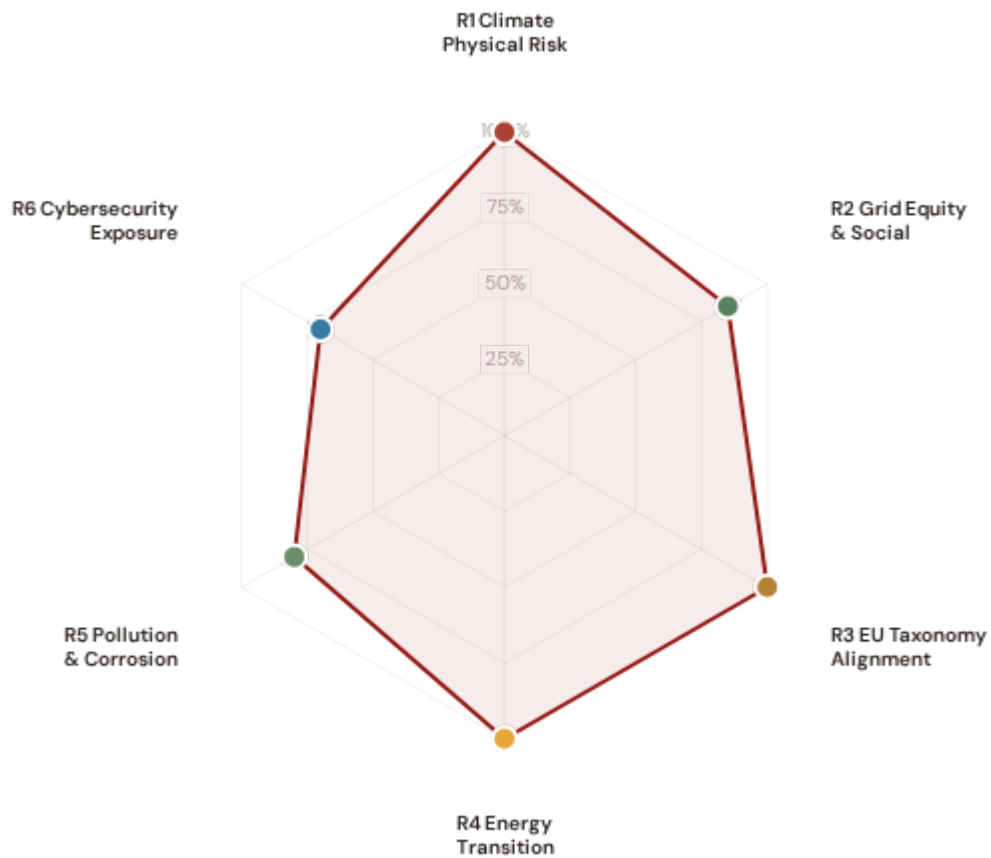
C3

ISO 9223 classification



ESG Radar — Six-Report Overview

Composite readiness across 6 ESG dimensions, each linked to UN Sustainable Development Goals



R1 · Climate Physical Risk Assessment **READY**

SDG 13 Climate Action · SDG 9 · SDG 11

SDG 13 **SDG 9** **SDG 11**

R2 · Grid Equity & Social Vulnerability **READY**

SDG 7 Affordable and Clean Energy · SDG 10 · SDG 1

SDG 7 **SDG 10** **SDG 1**

R3 · EU Taxonomy Alignment **READY**

SDG 9 Industry, Innovation, Infrastructure · SDG 13 · SDG 12

SDG 9 **SDG 13** **SDG 12**

R4 · Energy Transition & DER Stress **READY**

SDG 7 Affordable and Clean Energy · SDG 13 · SDG 9

SDG 7 **SDG 13** **SDG 9**

R5 · Pollution & Corrosion **READY**

SDG 11 Sustainable Cities and Communities · SDG 3 · SDG 15

SDG 11 **SDG 3** **SDG 15**

C

Audit Trail & Methodology Note

Traceability for CSRD Article 29a limited assurance

ITEM	VALUE
Scoring Engine	SSI Index v4.0.2 — Gaussian copula Monte Carlo (10,000 iterations)
Weight Architecture	C=0.30, V=0.10, I=0.25, E=0.10, S=0.20, T=0.05
Modifiers Applied	R3 C_mult: 1.061, R4 F_topo: 1.000, R6 restoration: 0.060, R6 seismic: 0.160, R6 volcanic: 0.100, R6 hurricane: 0.020, R6 hydro_deficit: 0.070, R7 cyber: 1.025
Degradation Model	Markov 5-state, CIGRE TB 761 calibration
Thermal Model	IEEE C57.91 transformer loading curves
Report Date	17 March 2026
Reporting Period	Calendar year 2025
Refresh Cadence	Annual (data ingestion → scoring engine → display)
Substation	ICE – Central Telefónica (w1230905408)
Data Assurance Level	All inputs from institutional open-source with citable vintage